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GBNet: Gradient Boosting packages integrated into PyTorch

Michael Horrell [1]

1 Independent Researcher, USA

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Software

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Summary

GBNet is a Python software package integrating the powerful Gradient Boosting Machines (GBMs) (Friedman, 2001) packages XGBoost (Chen & Guestrin, 2016) and LightGBM (Ke et al., 2017) with PyTorch (Paszke et al., 2019), a widely-used deep learning library. Gradient boosting is a popular machine learning technique known for accuracy in predictive modeling. XGBoost and LightGBM are industry standard implementations of GBMs recognized for their speed and strong performance across numerous applications (Kaggle, 2021). However, these libraries primarily handle standard machine learning tasks and are more difficult to use when faced with complex or non-standard modeling scenarios. For example, using non-standard loss functions with either XGBoost or LightGBM requires the user to manually compute gradients and hessians, a prohibitively difficult requirement for even moderately complex losses.

PyTorch is popular for its ease of defining and training neural networks. Its computational graph offers automatic differentiation capabilities. GBNet leverages these capabilities acting as the go-between that links gradients/hessian calculations from PyTorch to XGBoost or LightGBM models. This integration allows users to easily construct and train complex hybrid models that combine gradient boosting with neural network architectures. GBNet significantly broadens the scope of problems that can be solved with the world leading gradient boosting software packages.

Statement of need

While XGBoost and LightGBM are industry-standard solutions for tabular data machine learning problems, they have limited flexibility in defining complex model architectures tailored to specific problem types. Users wishing to define custom loss functions, novel architectures, or other advanced modeling scenarios face substantial difficulty to do so due to complex gradient and hessian calulations that are requirements to use either XGBoost or LightGBM.

As a simple motivating example, consider a forecasting model that has a linear trend and a periodic component. A natural specification of this model might be Forecast $(t) = t\beta +$ PeriodicFn (t), where β is a constant defining the trend and $PeriodicFn$ is modeled using a GBM. Despite its relative simplicity, this model cannot be easily fit using XGBoost or LightGBM alone.

GBNet addresses this limitation by providing PyTorch Modules that wrap XGBoost and LightGBM. These Modules are model building blocks like any other PyTorch Module. Valid code defining a PyTorch module accomplishing a version of Forecast (t) is given in just a few lines:

```
import torch
from gbnet.xgbmodule import XGBModule
```

```python
class ForecastModule(torch.nn.Module):
    def __init__(self, n, d):
        super().__init__()
        self.linear = torch.nn.Linear(d, 1)
        self.xgb = XGBModule(n, d, 1)

    def forward(self, t):
        return self.linear(t) + self.xgb(t)

    def gb_step(self):
        self.xgb.gb_step()
```

The new parts of this code (mainly in comparison to PyTorch code) are `XGBModule`, the wrapper on XGBoost and `gb_step`, a method used to update the underlying XGBoost model.

As seen in this example, once an instance of `XGBModule` has been defined, it can be combined with any other model logic supported by PyTorch. This straightforward example demonstrates GBNet's ease-of-use in defining complex models.

GBNet is the first software package to combine state-of-the-art gradient boosting software with neural network packages in a near-seamless and general way. Other packages either solve similar problems by providing Gradient Boosting packages with slightly more complex capabilities (Horrell (2020), Ingas (2024)) or, simply as a method to combine GBMs and Neural Networks, resort to different types of stacking or other more complex combinations (Kontschieder et al. (2015), Ke et al. (2019), Balestriero (2017), Popov et al. (2019)). GBNet allows users of the world's best gradient boosting packages to explore many of the rich architectural possibilities available through PyTorch.

Research Applications

Several research areas stand to benefit from GBNet. GBNet itself contains a forecasting application (`gbnet.models.forecasting`) that has improved performance over Meta's Prophet algorithm (Taylor & Letham, 2018) on a set of benchmarks seen in the notebook linked here. The package also provides an ordinal regression implementation (`gbnet.models.ordinal_regression`) featuring the ordinal loss which itself is complex, has fittable parameters and is not included in either XGBoost or LightGBM. A notebook here demonstrates the ordinal regression application.

More broadly, GBNet may benefit any researcher looking to leverage non-parametric methods while still retaining structural control over their model. In particular, researchers using PyTorch mainly for its ability to produce outputs suited for their application may prefer GBNet at times because XGBoost and LightGBM themselves are incredibly robust. Neural networks can be finicky, requiring many small adjustments and normalizations, while GBMs often just work.

Research into network architectures specifically tailored for GBMs may also hold intrinsic value. Several classic architectures previously explored exclusively with pure neural network methods are now accessible for GBMs through GBNet. Important concepts and methods such as embeddings (Mikolov et al., 2013), autoencoders (Hinton & Zemel, 1993), variational methods (Kingma et al., 2013), and contrastive learning (Hadsell et al., 2006) may exhibit novel and interesting properties when integrated with GBMs.

Software Description and Examples

GBNet comprises two primary submodules:

- `gbnet.xgbmodule`, `gbnet.lgbmodule`, `gbnet.gblinear`: Contain PyTorch Module classes (`XGBModule`, `LGBModule` and `GBLinear`) that integrate XGBoost, LightGBM and a linear

booster respectively.

- `gbnet.models`: Includes practical implementations of models using either `XGBModule` or `LGBModule`. Currently there are two implementations. `gbnet.models.forecasting` provides a Sci-kit Learn interface (Pedregosa et al., 2011) for an optimized version of Forecast (t) seen above. `gbnet.models.ordinal_regression` provides a Sci-kit Learn interface for Ordinal Regression.

Forecasting Example

`gbnet.models.forecasting.Forecast` is compared to the Meta Prophet algorithm over 500 independent trials as reported in the following table. Each trial consists of selecting a dataset uniformly at random, selecting a training cutoff uniformly at random, selecting a test period cutoff uniformly at random, and finally training a model and testing performance. The default `gbnet.models.forecasting.Forecast` beat Prophet in 74% of trials and had a higher than 50% win rate on 8 out of 9 datasets when comparing RMSE values. In addition, `gbnet.models.forecasting.Forecast`, when it did have the losing RMSE, tended to lose by less in comparison to Prophet.

Dataset	N trials	GBNet win Rate (%)	Avg. GBNet Losing RMSE Ratio	Avg. Prophet Losing RMSE Ratio
Air Passengers	50	**74%**	**1.42**	1.64
Pedestrians Covid	56	**66%**	**1.21**	1.73
Pedestrians Multivariate	54	**70%**	**1.34**	1.35
Retail Sales	75	**81%**	**1.26**	1.97
WP Log R	59	**90%**	**2.19**	2.60
WP Log R Outliers1	60	**77%**	**1.40**	2.56
WP Log R Outliers2	49	**71%**	**1.85**	2.47
WP Log Peyton Manning	45	44%	**1.36**	2.22
Yosemite Temps	52	**85%**	**2.16**	2.93

Code for these results is here.

Ordinal Regression Example

Ordinal regression requires fitting a cumulative logit model with breakpoints (McCullagh, 1980). The breakpoints are a set of parameters that, loosely speaking, separate the different ordinal classes. A plot showing the fitted probabilities on the Ailerons dataset from Gagolewski (2022) is below. Because the breakpoints were initialized to be evenly spaced, the different spacing seen in the figure illusrtates that `gbnet.models.ordinal_regression.GBOrd` fit the breakpoint parameters along with the underlying GBM.

Comparison of `gbnet.models.ordinal_regression.GBOrd` to alternatives across several datasets is linked here.

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